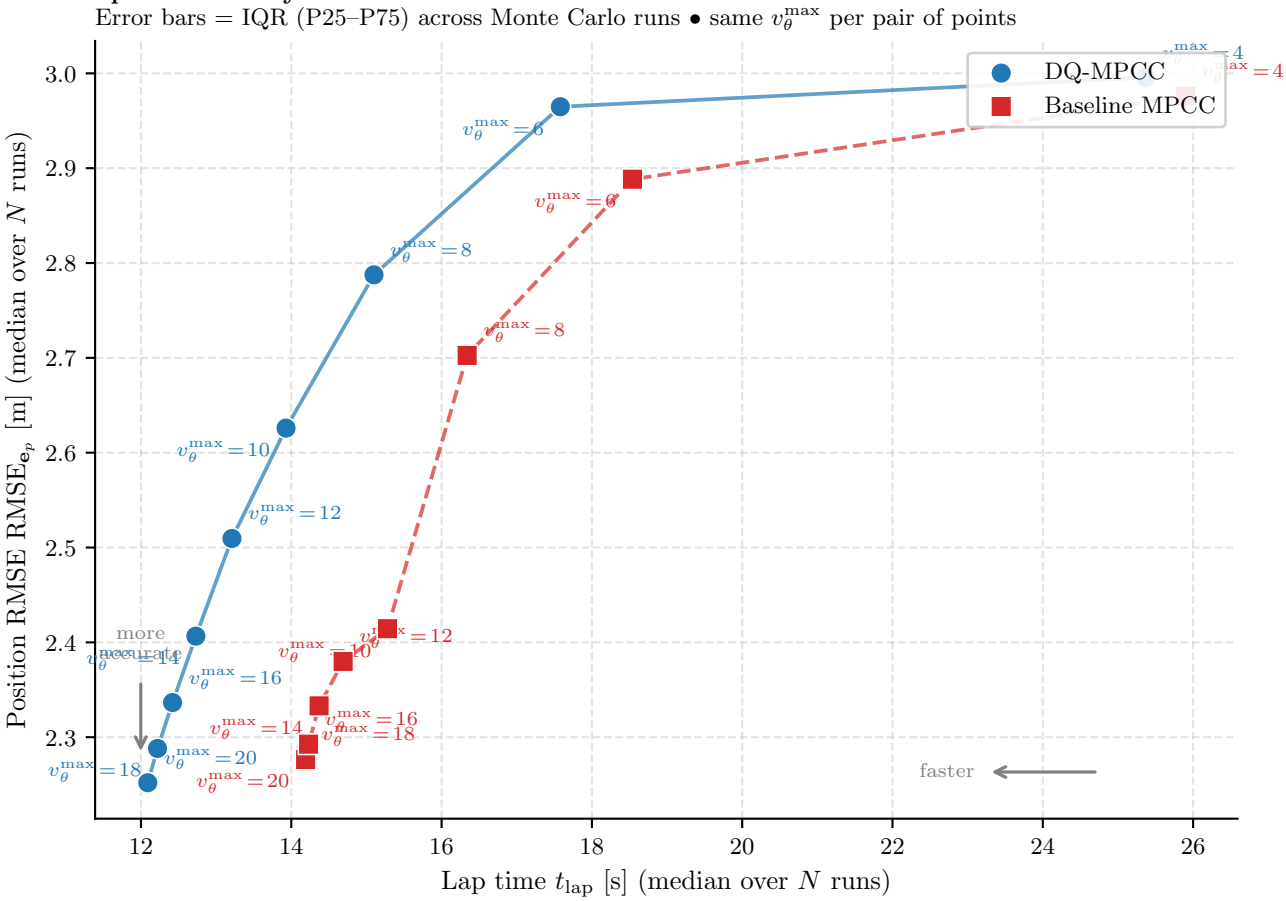
Speed–Accuracy Pareto Frontier
Error bars = IQR (P25–P75) across Monte Carlo runs • same $v_\theta^{\max}$ per pair of points
DQ-MPCC
Baseline MPCC
more accurate
faster
Position RMSE $\mathrm{RMSE}_{\mathbf{e}_p}$ [m] (median over N runs)
Lap time t_{lap} [s] (median over N runs)
3.0
2.9
2.8
2.7
2.6
2.5
2.4
2.3
12
14
16
18
20
22
24
26
$v_\theta^{\max}=4$
$v_\theta^{\max}=6$
$v_\theta^{\max}=8$
$v_\theta^{\max}=10$
$v_\theta^{\max}=12$
$v_\theta^{\max}=14$
$v_\theta^{\max}=16$
$v_\theta^{\max}=18$
$v_\theta^{\max}=20$